Modeling value based decision-making under stress and uncertainty

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Introduction

- In the real-world decisions are made incorporating latent information and variable environments, giving rise to uncertainty.
- Misestimation of uncertainty has been linked to various psychological disorders leading to characteristic changes in learning and explorative behaviour.
- Long-term exposure to stress is also known to increase the risk of mental health problems.
- We wanted to capture the impact of trait anxiety, acute stress on learning and decision making in a volatile environment through our study.

Methods

- We utilised a modified version of the restless bandit task used by Speekenbrink et al. (2015).
- There were four arms whose expected rewards followed the Gaussian random walks process.
- On pressing arm j, where j = {1,2,3,4}, the arm yielded a reward that was rounded off to the nearest integer.
- More volatile blocks had higher variance in the reward values of their arms.

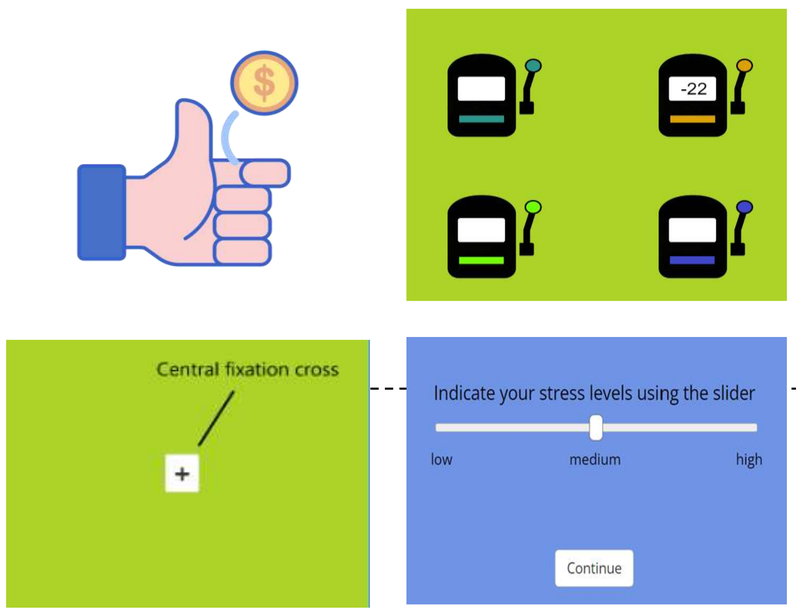

Figure A: The game consisted of 200 trials. Participants' monetary rewards were determined by their performance in the game. They were asked to report their stress levels after every 5 trials.

Data Analysis

- To understand the behavior of the participants over task conditions, we selected 2 metrics:
  - If the choice was **advantageous**, i.e., whether or not the arm with the best value was selected and,
  - If a **switch** from the previously chosen arm was made.
- We performed model agnostic analysis to inform policy of the agents.
- A binary dependent variable was assumed, for advantageous choice and switch, and Generalized Linear Mixed Models (GLMMs) were fit.

No independent variable
Volatility
Stress & Monetary loss
Learning across trials
Arm values
Previous rewards
Interaction effect between stability and arm values

Figure B: Conditions we tested for

- A total of 16 regression models were fit corresponding to different parameters. Model selection was done by the buildmer and GLMMadaptive packages in R.
- The best models for the two variables were-:

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advantageous ~ loss* volatility * stress + rewards + trial_no + (1 | ID) + arm values

switch ~ loss* volatility * stress + rewards + log(trial_no) + (1 | ID) + arm values
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Results

- The stress scores were significantly higher in the blocks where participants began with a loss (0.45) vs no monetary loss(0.32). The linear mixed model results were ($\beta = 0.11 \pm 0.019$, p =1.46 e-08).

For **Advantageous** choice:

- Odds of making an advantageous choice increased with decrease in volatility.

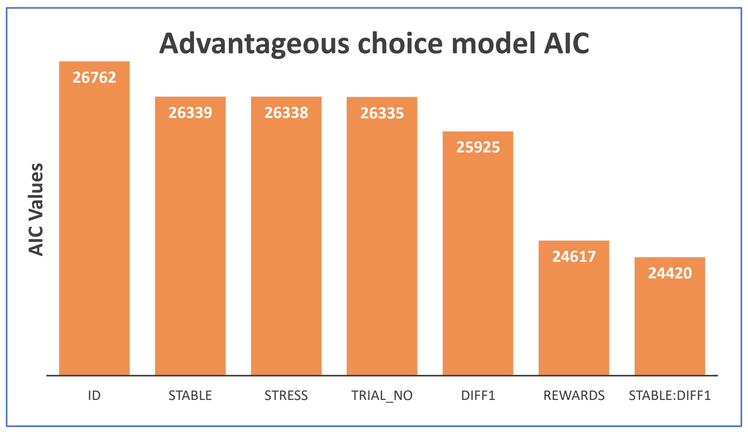

Figure C: Bar graphs illustrating the AIC values of different advantageous choice models. The first model corresponds to the base model and the x axis contains each new predictor added to the model. Diff1-: difference between best and second best arm

- The rewards that the participants had obtained in the previous trials also impacted their performance in subsequent trials.
- Subjects that began with a **monetary loss** were more likely to make optimal choices.
- Subject performance improved across trials.

For **Switching** behaviour-:

- In stable blocks, the number of switches were much fewer for the subjects.
- If the participants had obtained a high positive reward in the previous trials, they were unlikely to switch to a different arm in the next trial.

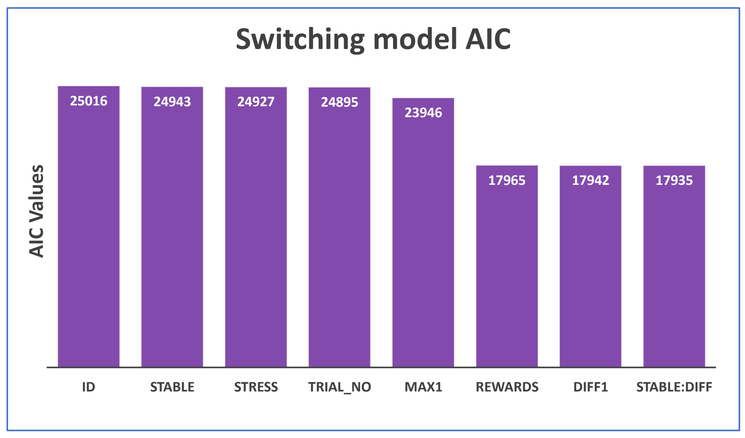

Figure D: Bar graphs illustrating the AIC values of different switching models. The first model corresponds to the base model and the x axis contains each new predictor added to the model.

- The players explored less with more trials
- Higher stress scores correlated with less exploration as participants deviated little from their choices

Future Directions

- Identifying physiological and neural correlates underlying stress response and uncertainty estimation of the decision making of the participants.
- Modeling subject behaviour by **Hierarchical Drift Diffusion Models(HDDMs)** to incorporate internal representations and **latent information** such as information accumulation rates and choice biases.
- Planning to build RL agents to understand learning and decision making in these conditions

References

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2. Speekenbrink M, Konstantinidis E. Uncertainty and exploration in a restless bandit problem. Top Cogn Sci. 2015. Apr;7(2):351-67. doi: 10.1111/tops.12145. Epub 2015 Apr 20. PMID: 25899069.
3. 3. Aylward, J., Valton, V., Ahn, WY. et al. Altered learning under uncertainty in unmedicated mood and anxiety disorders. Nat Hum Behav 3, 1116–1123 (2019). https://doi.org/10.1038/s41562-019-0628-0.